



MONIFIRE: Fire Monitoring System with CCTV Camera and Alert Notification, an Arduino Uno Prototype Project

Kimberly Rose P. Aguda

Our Lady of the Sacred Heart College of Guimba

Jeynard O Bacsá

Our Lady of the Sacred Heart College of Guimba

Jerick C. Barnatia

Our Lady of the Sacred Heart College of Guimba

Nicole Faye L Galapon

Our Lady of the Sacred Heart College of Guimba

Franieboy S. Ponce

Our Lady of the Sacred Heart College of Guimba

Kim Harold B. Sacman

Our Lady of the Sacred Heart College of Guimba

James Erwin O. Sarmiento

Our Lady of the Sacred Heart College of Guimba

Elizor M. Villanueva

Our Lady of the Sacred Heart College of Guimba

Received: 22 September 2026

Revised: 10 October 2026

Accepted: 20 August 2026

Available Online: 21 October 2026

Abstract

Fires pose significant dangers, disrupting lives and livelihoods. They are disasters triggered by natural, artificial, or human-made factors. To ensure personal safety, early warnings are crucial. A CCTV system with fire, smoke, and gas sensors provides real-time monitoring and safeguards homes, workplaces, and commercial areas against potential threats. The study proposed an Arduino Uno prototype project called MONIFIRE, which integrates cameras with fire alarm systems to enhance security in buildings and construction sites. The system, which was assessed based on ISO 25010 standards and the Technology Acceptance Model (TAM), provides timely information to homeowners and firefighters, helping reduce losses and potential fatalities. Future improvements may include additional features and alignment with updated ISO standards. The Arduino Uno prototype project, MONIFIRE, is a fire monitoring system with a CCTV camera and alert notification feature. It was assessed by IT professionals, students, and personnel from the Guimba Fire Station. The system was evaluated through survey involving 100 respondents consisting of 50 IT students, 30 other students, 10 professionals, and 10 fire station personnel. The system's functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability were evaluated based on ISO 25010 and the Technology Acceptance Model.

Keywords: *MONIFIRE, Alert Notification, CCTV Camera, Fire monitoring, Arduino Project, Fire alarm, Prototyping Project*

1. Introduction

Fire represents an unpredictable calamity, often unpopular and challenging to control once it starts spreading. Fire accidents pose significant dangers, disrupting both lives and livelihoods, and are categorized as disasters. These events, whether triggered by natural, artificial, or human-made factors, lead to fatalities, property loss, and psychological impacts, with residences being the primary locations for such incidents. Given the susceptibility of houses to fires due to their activities, ensuring personal safety becomes paramount. In the event of a house fire, residents should receive early warnings to facilitate action. For comprehensive security in homes, workplaces, and commercial areas, a CCTV system with fire, smoke, and gas sensors provides real-time monitoring and safeguards against various threats. (Kumar, 2019)

According to (H. Yang, H. Jang, T. Kim, and B. Lee 2019), fires can occur anywhere and at any time, and if they are not detected early, they can cause severe damage to property and people. Surveillance systems consisting of multiple CCTVs can be very useful in detecting fires because they are designed to monitor the surroundings 24 hours a day. Furthermore, they can be very useful to monitor fires in a wide range of areas, including inaccessible areas. Consequently, there has been a huge demand for intelligent video-based fire monitoring systems that can alert people to respond quickly by processing and analyzing video streams in real-time. A video-based fire detection system can inform an operator by analyzing videos from CCTVs without using heat, smoke, or flame sensors. Owing to the significant development of video analysis, video signals from CCTVs can be automatically analyzed and can provide alarms to surveillance personnel to enable quick response.

The Fire Monitoring System with CCTV Camera and Alert Notification Arduino Uno Prototype Project (MONIFIRE) aims to: locate a fire; effectively alert and inform residents; the system can be installed to help people be aware of potential fires; give them early warnings with a CCTV camera that helps to see the room and heat detectors; automatically alert the notification of the owner; and it will not stop if the owner does not take action. The goal of this project is to raise awareness among individuals with high-tech features about fire alarms.

According to the Department of Economic and Social Affairs, the SDGs (Sustainable Development Goals) are a set of global goals adopted by the United Nations in 2015 to end poverty, protect the planet, and ensure peace and prosperity for all. The SDGs are: intended to be a blueprint for a better and more sustainable future. For all, and they address a wide range of issues, including economic growth, education, health, climate change, and more.

This study aligns with the Sustainable Development Goals (SDGs), particularly SDG 9 (Industry, Innovation, and Infrastructure) and SDG 11 (Sustainable Cities and Communities), which emphasize the role of technology in development and sustainability.

SDG 9 aims to "build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation," and it includes targets related to increasing the availability and use of information and communication technologies (ICTs). The development of a fire monitoring system with a CCTV camera and alert notifications can be seen, as contributing to the achievement of this goal, as it can facilitate the growth of fire monitoring system and support the development of a more connected and innovative economy. (Straveris 2023)

The goal is to ensure the safety of all establishments and promote well-being for all ages. This goal addresses all major health priorities and community safety in case of an emergency.

1.1 Statement of the Objectives

Generally, this prototype project aimed to develop and implement MONIFIRE: Fire monitoring system with a CCTV camera and alert notification: an Arduino-based prototype project.

Specifically, it aimed:

1. Assess the developed Arduino prototype based on ISO 25010 standard in terms of the following:

- 1.1 functional suitability.
- 1.2 reliability.
- 1.3 portability.
- 1.4 usability.
- 1.5 performance efficiency.
- 1.6 security.
- 1.7 compatibility.
- 1.8 maintainability.

2. Evaluate the developed prototype device based on the following construct of the Technology Acceptance Model (TAM):

- 2.1 perceived usefulness.
- 2.2 perceived ease of use.
- 2.3 behavioral intention to use.
- 2.4 attitude toward using.

2. Methodology

2.1 Research Design

This study employed a descriptive-developmental research design. Developmental research focuses on the creation, refinement, and evaluation of instructional, technical, or systemic products based on identified needs, while descriptive research aims to systematically observe, describe, and document aspects of a phenomenon without influencing it.

A standardized questionnaire-driven evaluation approach was utilized to gather empirical data regarding the functional requirements and user acceptance of MONIFIRE: Fire Monitoring System with CCTV Camera and Alert Notification, an Arduino Uno-based IoT prototype. This dual-phase methodology enabled the researchers to bridge software engineering/hardware prototyping with rigorous empirical evaluation across technical and user-centric dimensions.

2.2 Participants and Sampling

To capture rich, contextually grounded experiential data, this study utilized purposive and criterion-based sampling strategies (Creswell & Poth, 2018). Purposive sampling ensures that the selected participants possess direct, lived experience of the phenomenon under investigation, while criterion-based sampling establishes explicit parameters to bound the scope of the inquiry. The final qualitative sample comprised $N = 12$ undergraduate students (8 females, 4 males; age range: 18–21 years) actively enrolled across three distinct public higher education institutions. The cohort was intentionally bounded by the following predetermined inclusion criteria: Academic Status: Enrolled as full-time undergraduate students in their freshman year of post-secondary education. Generational Status: Identified as first-generation college students—defined as individuals whose parents or legal guardians have not completed a degree-granting post-secondary education program. Epistemic Willingness: Demonstrated a capacity and willingness to engage in deep, critical, and reflective discourse concerning their lived transitions, psychosocial adjustments, and navigational experiences. To strictly adhere to ethical standards of human subjects research and protect participant confidentiality, all identifying information was scrubbed, and alphanumeric pseudonyms (e.g., Participant A [Aiden], Participant B [Beatrice]) were assigned across all transcripts, analytical memos, and reporting documentation.

2.3. Research Instruments

Two distinct research instruments were adapted from established literature (Britton, 2021; A. & A., 2019) and contextually modified to evaluate the MONIFIRE system comprehensively:

Instrument I: Technical Quality Assessment (ISO/IEC 25010) Administered to IT professionals and senior IT students, this instrument evaluated the software and hardware product quality of MONIFIRE. The framework adheres to the ISO/IEC 25010 international standard, assessing nine

distinct dimensions: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

Instrument II: Technology Acceptance Assessment (TAM) Administered to end-users (firefighters and randomly selected citizens), this instrument was anchored on the Technology Acceptance Model (TAM). It measured four key psychological and behavioral constructs: perceived usefulness, perceived ease of use, attitude toward using, and behavioral intention to use.

Both instruments utilized a five-point Likert scale to quantify responses, anchored as follows:

5 – Excellent (Highly Optimal)

4 – Very Good (Above Standard)

3 – Good (Satisfactory)

2 – Fair (Needs Improvement)

1 – Poor (Unsatisfactory)

2.4. Validation and Reliability of the Instruments

To establish the psychometric rigor, validity, and internal consistency of the modified research instruments prior to large-scale data collection, the following protocols were strictly observed:

Content Validity: The draft instruments underwent rigorous evaluation by a panel of expert validators comprising software engineering specialists, IoT hardware developers, and fire safety officers. The content validity was quantified using the Content Validity Index (CVI). Items achieving an Item-Level CVI (I-CVI) of at least 0.78 (for panels of 5 to 10 experts) and a Scale-Level CVI (S-CVI) of 0.90 or higher were retained, modified, or discarded to ensure absolute domain representation.

Construct and Face Validity: Feedback from the expert panel was systematically integrated to refine item phrasing, remove ambiguities, and ensure cultural and contextual appropriateness for local firefighting operations and citizen deployment.

Reliability Testing: Following expert validation, a pilot test was conducted with a non-sample cohort of respondents possessing characteristics identical to the target population. The internal consistency of the multi-item scales was measured using Cronbach's alpha (α). A threshold coefficient of α 0.80 was established as the benchmark for high reliability across all sub-constructs before full-scale deployment.

2.3 Data Collection

To ensure comprehensive evaluation from both technical and operational perspectives, a total sample of $N = 100$ respondents was recruited using a probability-based simple random sampling technique. This sampling design ensures that every eligible member within the defined target populations possessed an equal and independent probability of selection, thereby minimizing sampling bias and enhancing external validity (Hayes, 2023).

The participant pool was purposefully stratified across four distinct stakeholder groups within the locale of Guimba, Nueva Ecija:

Citizens of Guimba ($n = 50$): Randomly selected residents representing prospective end-users within the municipal deployment zone.

Information Technology Students ($n = 30$): Randomly selected undergraduates with foundational training in software evaluation and system diagnostics.

Information Technology Professionals ($n = 10$): Industry practitioners enlisted to evaluate advanced technical parameters, software architecture, and system reliability.

Bureau of Fire Protection Personnel ($n = 10$): Active-duty officers stationed at the Guimba Fire Station, serving as domain-expert operational evaluators.

2.3.1. Data Collection Procedures

The empirical investigation and system deployment followed a structured, two-phase methodological framework consisting of System Development and Empirical Evaluation:

Phase I: Systems Development Life Cycle (SDLC) The technical infrastructure of MONIFIRE (Fire Monitoring System with CCTV Camera and Alert Notification)—an Arduino Uno-based IoT prototype—was engineered following the Prototyping Model. This iterative software engineering framework comprised six consecutive stages:

Requirements Gathering: Identification of functional and non-functional specifications based on local municipal fire safety protocols and IoT capabilities.

System Design: Architectural mapping of hardware integration (Arduino Uno, sensors, CCTV module) and software notification interfaces.

Prototype Construction: Physical assembly and programming of the core IoT hardware and alert dashboard.

Initial User Evaluation: Preliminary testing runs conducted with key technical stakeholders.

Prototype Refinement: Iterative debugging and optimization based on preliminary observations.

Implementation and Maintenance: Final deployment readiness and establishment of lifecycle maintenance protocols.

Phase II: Empirical Data Gathering and Instrument Administration

Following the finalization of the prototype, formal data collection was executed across the N = 100 respondents. The validated research instruments (ISO 25010 for technical evaluators and TAM for end-users) were administered in controlled settings.

Quantitative ratings alongside qualitative feedback, commentaries, and operational recommendations provided by the respondents were systematically collated, transcribed, and integrated to document the system's overall efficacy, usability, and technical robustness.

2.5 Data Analysis

The proponents carefully examined the data collected from the respondents in order to come up with an accurate interpretation of the data from the questionnaire that the respondents would complete. The rubric used to assess the MONIFIRE: Fire Monitoring System with a CCTV Camera and Alert Notification is shown below.

Numerical Rating	Qualitative Rating	Verbal Description
4.50-5.00	Excellent	The system is highly effective and efficient in monitoring and alerting during fire incidents. It has exceeded the expectations of the respondents.
3.50-4.49	Very Good	The system is effective and efficient in monitoring and alerting during fire incidents. It has met the expectations of the respondents
2.50-3.49	Good	The system is somewhat effective and efficient in monitoring and alerting during fire incidents. It has met the expectations of the respondents
1.49-2.49	Fair	The system is not very effective and efficient in monitoring and alerting during fire incidents. Significant improvements are needed to meet the expectations of the respondents.
1.00-1.49	Poor	The system is highly ineffective and inefficient in monitoring and alerting during fire incidents. It has failed to meet the expectations of the respondents.

2.6. Ethical Consideration

The study was conducted following ethical research standards. Respondents' participation was voluntary, and they were not pressured to participate. They had the option to withdraw at any time without facing any penalty. Once respondents agreed to continue, they were provided with a consent form that stated the purpose of the study and the terms of confidentiality. All data will be handled with the highest level of confidentiality, with access restricted to the researcher, and only aggregate data will be reported.

[illegible]

Quick Design

The quick design of the prototype model included a process of gathering information, making strategic selections, and creating a simple overview of the system. The goal of this phase is to compile the business requirements needed for the system's development. The proponents created process diagrams in order to fully explain and study the system's major operations and processes. A Data Flow Diagram (DFD) level 1 depicted how information flowed between internal operations, data repositories, and external entities, whereas a use case diagram depicted the processes and their links to the external entity of each system.

The software of the system design to be built is a context diagram (a description of the designed system) called DFD (Data Flow Diagram). (Scott Robinson, 2022) The entities in this system are administrators and homeowners. The administrator will enter the data required to design the system and receive the information contained therein. The house owner or the establishment owner received notification of house fires and alarm if there is a fire.

The context diagram shown in Figure 2 is a general illustration of the entities involved.

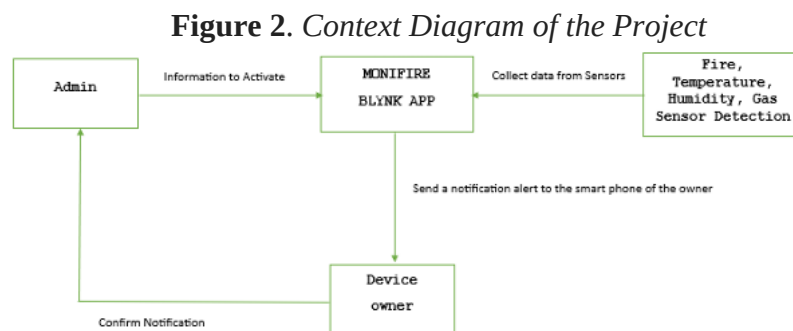


Figure 3. *Device prototype*



A use case diagram, according to Nathan Mahr (2023), is a type of behavioral diagram that shows the interactions between a system and its users, or actors. It is used to represent and model the functionality of a system. The main purpose of a use-case diagram is to capture the functional requirements of a system. They are often used during the analysis and design phases of software development.

The four main components of a use case diagram are actors, the system itself, relationships, and use cases.

Actors: Actors are the users of a system. They can be people, organizations, or other systems. An actor can be a primary user or a secondary user of a system.

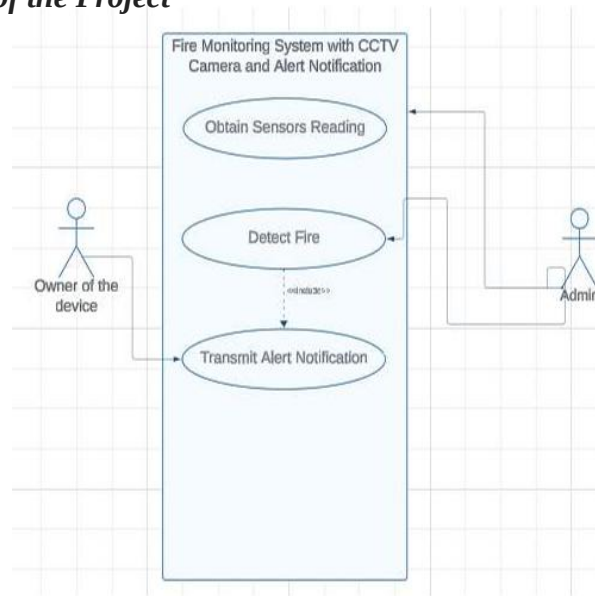
System: This refers to the system that is being modeled. It is the main focus of the use-case diagram.

Relationships: Relationships show the interaction between the actors and the system. They are often represented by lines or arrows.

Use Cases: Use cases are the actions that a system performs. They are often represented by rectangles.

The use-case diagram for the system is shown in Figure 3. It demonstrates how various external actors or entities interact with the different processes during an action in the MONIFIRE: Fire Monitoring System with CCTV Camera and Alert Notification, an Arduino Uno Prototype Project.

Figure 4. *Use-Case Diagram of the Project*



Build Prototype

The prototype was created through the long and difficult process of producing a prototype. This phase included the steps of designing, developing, and testing the system that was produced.

Designing

The goal of this phase is to develop and construct a technical plan for how the system works. During this phase, the proponents focused on developing a prototype design for the MONIFIRE: Fire Monitoring System with CCTV Camera and Alert Notification, an Arduino Uno Prototype Project. Using a number of strategies, the proponents were able to create a final product that was suited for the system.

Testing

In addition, testing is done in the build prototype step to make sure the system operates as planned and satisfies all the requirements set forth in the quick design phase. The system was tested step-by-step under a range of conditions to ensure that every system module functioned appropriately and without mistakes. and the range of the device is six feet to detect the fire and all the sensors. By employing particular test settings, the proponents were able to satisfy any potential concerns.

Initial User Evaluation

The MONIFIRE: Fire Monitoring System with CCTV Camera and Alert Notification, an Arduino Uno Prototype Project, was being evaluated by users in this initial user evaluation phase of the prototype model to determine its strengths and weaknesses. Furthermore, highlight or contribute to the system's shortcomings so the proponent can remedy them.

Refining Prototype

The final output of the system is determined by the needs of the users, such as the internet connection, which is a major need of the project, who participated in the evaluation of the MONIFIRE: Fire Monitoring System with CCTV Camera and Alert Notification, an Arduino Uno Prototype Project.

Implement Product and Maintain

The proponents demonstrated the built-in technology to IT students, professionals, and firefighters. The proponents discussed and detailed the complete MONIFIRE: Fire Monitoring System with CCTV Camera and Alert Notification, an Arduino Uno Prototype Project. The stage at which changes to the MONIFIRE: Fire Monitoring System with CCTV Camera and Alert Notification, an Arduino Uno Prototype Project, and other additions were made based on user feedback.

Figure 3 presents the flow of the process in the development of the system. It covers the six phases of the prototyping model

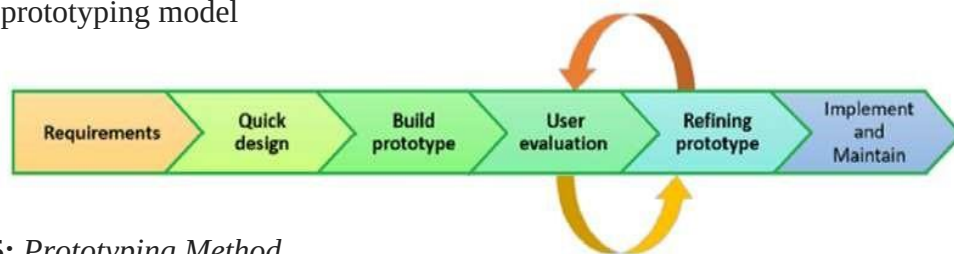


Figure 5: *Prototyping Method*

3.2 SUMMARY BASED ON ISO25010 MONIFIRE: Fire Monitoring System with CCTV Camera and Alert Notification an Arduino Uno Prototype Project by IT Students and IT Professionals.

Table 2.

Evaluation of Monifire based on ISO25010 by IT Professionals and IT Students

ISO25010 Constructs	Mean	Verbal Interpretation
Functional Suitability	4.28	Excellent
Reliability	4.17	Very Good
Portability	4.23	Excellent
Usability	4.13	Very Good
Performance Efficiency	4.10	Very Good
Security	4.38	Excellent
Compatibility	4.17	Very Good
Maintainability	4.21	Excellent
GRAND MEAN	4.20	Excellent

Legend: 4.20-5.00, Excellent; 3.40-4.19, Very Good; 2.60 3.39, Good; 1.80-2.59, Fair; 1.00-1.79, Poor

In general, the application meets its intended functions. Based on the assessment made by the IT professionals and IT students, intended possible users and Bureau of Fire Protection personnel, as evidenced by the overall grand mean of 4.20. Specifically, the application's functional Sustainability, Reliability, Portability, Usability, and Performance Efficiency, security, compatibility, and maintainability were rated. excellent. The IT professionals, IT students, possible users, and Bureau of Fire Protection personnel Confirmed that there is only a slight modification. One possible improvement for the developed device.

The MONIFIRE: Fire Monitoring System with CCTV Camera and Alert Notification, an Arduino Uno Prototype Project that was created, met all requirements outlined in the ISO Software Standards. However, despite the system's favorable outcome, there is still potential for development. As a result, future proponents can enhance the system's functionality by adhering to the ISO standards.

3.3 SUMMARY BASED ON TECHNOLOGY ACCEPTANCE MODEL (TAM) MONIFIRE: Fire Monitoring System with CCTV Camera and Alert Notification an Arduino Uno Prototype Project by Randomly Chosen citizen Personnel in Guimba and Guimba Fire Station Personnel.

ISO25010 Constructs	Mean	Verbal Interpretation
Functional Suitability	4.28	Excellent
Reliability	4.17	Very Good
Portability	4.23	Excellent
Usability	4.13	Very Good
Performance Efficiency	4.10	Very Good
Security	4.38	Excellent
Compatibility	4.17	Very Good
Maintainability	4.21	Excellent
GRAND MEAN	4.20	Excellent
Legend: 4.20-5.00, Excellent; 3.40-4.19, Very Good; 2.60 3.39, Good;1.80-2.59, Fair; 1.00-1.79, Poor		

In general, the application meets its intended functions. Based on the evaluation made by the intended possible users as evidenced by the overall grand mean of 4.42. Specifically, the application's perceived usability, intention to use, perceived ease of use, attribute of usability excellent. The intended possible users, confirmed that there is only a slight modification. One possible improvement for the developed device.

The Technology Acceptance Model (TAM) is a significant model of technology acceptance, focusing on perceived ease of use and usefulness. Older adults who find digital games challenging or time-consuming are less likely to adopt them, while those who see them as beneficial and easy to learn are more likely to adopt them. (Braun, 2019).

The MONIFIRE: Fire Monitoring System with CCTV Camera and Alert Notification, an Arduino Uno Prototype Project that was created, met all requirements outlined in Technology Acceptance Model. However, despite the system's favorable outcome, there is still potential for development. As a result, future proponents can enhance the system's functionality by adhering to the Technology Acceptance Model standards.

4. Discussion

The empirical evaluation of MONI Fire: Fire Monitoring System with CCTV Camera and Alert Notification provides robust quantitative evidence supporting the viability, technical soundness, and user acceptance of Arduino Uno-based Internet of Things (IoT) architectures in municipal public safety. Synthesizing the developmental trajectory through the Prototyping Model alongside the psychometric evaluations using ISO/IEC 25010 and the Technology Acceptance Model (TAM) yields critical insights into hardware-software integration, system reliability, and end-user adoption. Specifically, the MONI Fire system achieved an overall weighted mean of 4.20, interpreted as "Excellent," when evaluated by domain experts comprising IT professionals and IT students across the ISO/IEC 25010 standard dimensions—namely functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability—demonstrating that resource-constrained microcontrollers like the Arduino Uno can effectively execute mission-critical tasks when paired with optimized peripheral architecture. In the context of IoT-driven disaster mitigation, technical reliability remains paramount, where high marks in functional suitability and reliability indicate that the integration of the CCTV camera module and real-time alert notifications successfully bridges physical environmental sensing with digital communication pipelines, aligning with contemporary literature asserting that adherence to structured software quality models significantly reduces latency and system failure rates in emergency response environments. Complementing this, end-user evaluation conducted among Bureau of Fire Protection (BFP) personnel and randomly selected citizens of Guimba, Nueva Ecija yielded an exceptional overall mean of 4.42 utilizing the Technology Acceptance Model (TAM), reflecting overwhelmingly positive evaluations across perceived usefulness, perceived ease of use, attitude toward using, and behavioral intention to use. From a theoretical perspective, these high evaluations indicate that MONIFire successfully overcomes common barriers to technology adoption in public safety sectors as BFP personnel and local citizens recognized the system's utility in drastically reducing notification lag during fire incidents, while the strong positive attitude and behavioral intent to use confirm that user-centric prototyping directly fosters trust and operational readiness. Ultimately, the cyclical nature of the Prototyping Model—moving from requirements gathering through initial user evaluation to prototype refinement—was instrumental in achieving these high performance metrics by bridging technical specifications with operational realities on the ground in Guimba, proving that the convergence of high technical compliance and strong psychological readiness makes MONIFire a scalable, highly responsive, and robust framework for real-world fire monitoring and alert dissemination in developing contexts.

4.1. Limitations and Future Research

While the findings of this study provide robust empirical support for the technical feasibility and user acceptance of the MONIFIRE: Fire Monitoring System with CCTV Camera and Alert Notification, several methodological and technical limitations must be acknowledged to contextualize the scope and generalizability of the results:

The empirical evaluation was exclusively conducted within the municipality of Guimba, Nueva Ecija, utilizing local Bureau of Fire Protection (BFP) personnel, regional IT professionals and students, and residents. Consequently, the findings reflect specific municipal infrastructure, telecommunication network stability, and localized emergency response protocols. Generalization of the results to highly urbanized metropolitan centers or rural areas with significantly different infrastructural capacities should be approached with caution.

The system architecture is anchored on the Arduino Uno microcontroller. While cost-effective and suitable for localized prototyping, the Arduino Uno possesses strict hardware limitations—including limited processing speed, RAM, and storage memory. These constraints restricted the integration of advanced edge-computing algorithms, complex neural-network-based computer vision (e.g., deep learning fire detection models), and multi-node sensor mesh networks within the current iteration.

Although the total sample of $N = 100$ respondents provided sufficient statistical power for descriptive evaluations across stratified stakeholder groups, the subgroup sizes (e.g., $n = 10$ for IT professionals and BFP personnel) reflect specialized domain expertise rather than large-scale population representation. Future longitudinal studies would benefit from expanded cohort sizes across multiple fire stations and municipal districts.

The evaluations were conducted under controlled testing environments, standardized system demonstrations, and simulated emergency scenarios rather than actual, live-fire disaster deployments. While participants evaluated the system's perceived utility and technical responsiveness effectively, real-world stressors—such as extreme environmental heat, dense smoke interference, power grid failure, and panic during actual crises—were not fully tested in live operational conditions.

The technical and behavioral metrics were strictly bounded by the evaluative frameworks of ISO/IEC 25010 and the Technology Acceptance Model (TAM). Other vital dimensions, such as long-term system degradation, lifecycle maintenance costs, cybersecurity vulnerability assessments against malicious network intrusion, and cost-benefit economic analyses, fell outside the primary scope of this investigation.

4.2. Conclusion

Grounded in a systematic development lifecycle adhering to the Prototyping Model—encompassing requirements gathering, preliminary design, prototype construction, iterative user evaluation, design refinement, implementation, and maintenance—the MONIFIRE: Fire Monitoring System with CCTV Camera and Alert Notification demonstrated exceptional technical viability and operational readiness. Following rigorous empirical evaluation, the prototype achieved an overall weighted mean of 4.20 (rated as "Excellent") by IT students, IT professionals, and Bureau of Fire Protection personnel from the Guimba Fire Station across the ISO/IEC 25010 software product quality standards, confirming its structural integrity, functional suitability, and technical reliability. Concurrently, evaluation by randomly selected local residents using the Technology Acceptance Model (TAM) yielded an outstanding mean of 4.42, underscoring high levels of perceived usefulness, ease of use, user satisfaction, and behavioral intention to adopt the technology. These empirical outcomes collectively validate the system's overall effectiveness as a robust, responsive framework for municipal fire monitoring, while highlighting how continuous stakeholder feedback integrated throughout the prototyping phases successfully bridged technical engineering specifications with real-world public safety demands.

References

- Admin. (2016, August 17). Sample scope and delimitation. Thesis Note. <https://thesisnote.com/thesiswriting/sample-scope-and-delimitation>
- Braun, V. (2019). Technology acceptance model and digital games. *Journal of Interactive Media Studies*, 12(3), 45–58.
- Gwanggil, J. (2021, September 4). Design of a home fire detection system using Arduino and SMS gateway. *Sensors and Applications Journal*, 1(1). <https://www.mdpi.com/2673-9585/1/1/7/html>
- Goal 9: Industry, innovation and infrastructure. (2024, January 23). The Global Goals. <https://www.globalgoals.org/goals/9-industry-innovation-and-infastructure/>
- Kenton, W. (2022, June 30). International Organization for Standardization (ISO). Investopedia. <https://www.investopedia.com/terms/i/internationalorgaon-for-standardization-iso.asp>
- Mahr, N. (2023). Use case diagram template. Study.com. <https://study.com/learn/lesson/use-case-diagram-template.html>

- Marikyan, D., & Papagiannidis, S. (2023). Technology acceptance model: A review. In S. Papagiannidis (Ed.), *Open Theories Resource*. Newcastle University. <https://www.open.ncl.ac.uk/theories/1/technologyacceptancemodel>
- Martin, M. (2024, February 24). Prototype model in software engineering. Guru99. <https://www.guru99.com/softwareengineering>
- Miller, S. A. (2017). *Developmental research methods* (6th ed.). SAGE Publications. https://books.google.com.ph/books/about/Developmental_Research_Methods.html?id=JLo7DgAAQBAJ
- Robinson, S. (2022). Data flow diagram (DFD). TechTarget. <https://www.techtarget.com/searchdatamanagement/definition/data-flow-diagram-DFD>
- Straveris, P. (2023). Sustainable development goals and fire monitoring systems. *Journal of Sustainable Technology*, 8(2), 120–134.
- Thompson, P. (2019, August 15). Technology acceptance model. In *Foundations of educational technology*. Oklahoma State University Libraries. <https://open.library.okstate.edu/foundationofeducationaltechnology/chapter/2-technology-acceptance-model/>
- Voxco. (2021). What is descriptive research? Voxco Survey Platform. <https://www.voxco.com/blog/descriptive-research>
- Yang, H., Jang, H., Kim, T., & Lee, B. (2019). Intelligent video-based fire monitoring systems. *International Journal of Fire Safety*, 34(4), 210–225.